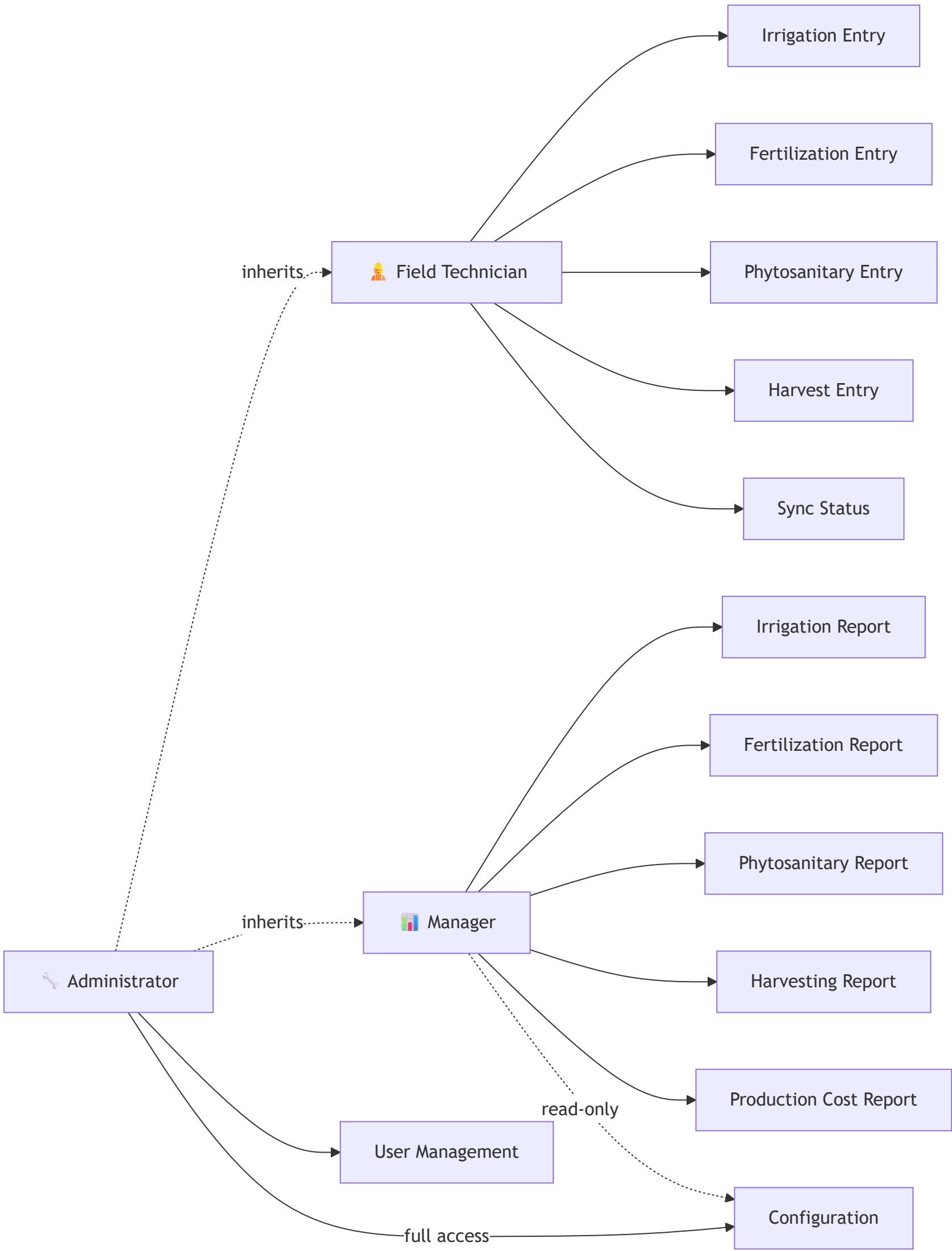


Agri-Sync — Features Specification

Overview

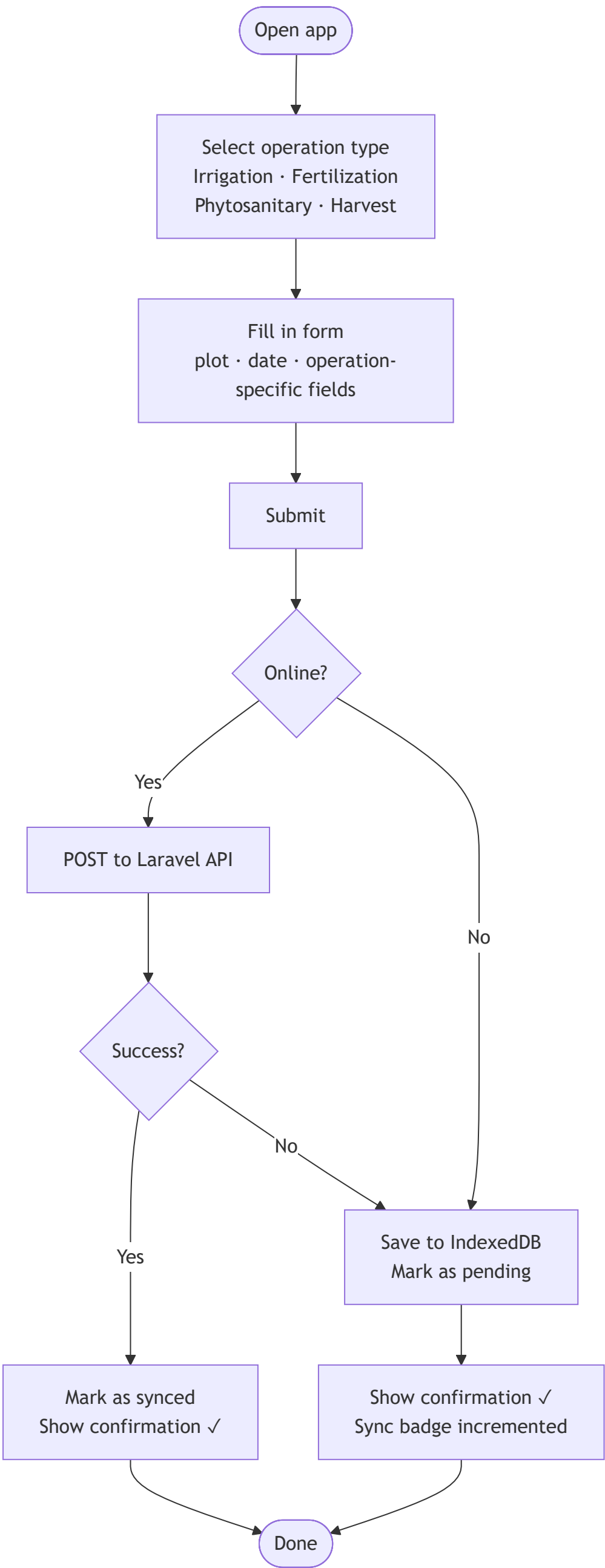
Agri-Sync is organised into four functional areas:

1. **Data Entry** — used by the Field Technician via the mobile PWA
2. **Reporting & Analytics** — used by the Manager via the web dashboard
3. **Configuration & Settings** — used by the Administrator via the web dashboard
4. **User Management** — accessible only to the Administrator



1. Data Entry (Field Technician — Mobile PWA)

The technician records farming operations per plot. All four operation types follow the same workflow: fill in the form, submit, and the record is saved — immediately if online, or queued for sync if offline.



1.1 Irrigation

Records water applied to a plot on a given date.

Field	Type	Notes
Plot	Select	Chosen from the configured plot list
Date	Date picker	Date of the irrigation event
Water quantity	Number	In the configured measurement unit (e.g. m³, litres)

1.2 Fertilization

Records a fertilizer application, including which product was used.

Field	Type	Notes
Plot	Select	Chosen from the configured plot list
Date	Date picker	Date of the fertilization event
Fertilizer	Select	Chosen from the configured fertilizer list
Quantity applied	Number	In the unit defined for that fertilizer

The system automatically derives NPK values from the quantity and the fertilizer's configured N/P/K percentages. No manual calculation is required from the technician.

1.3 Phytosanitary Treatments (Pesticides)

Records a pesticide application, including the target pest and any observations.

Field	Type	Notes
Plot	Select	Chosen from the configured plot list
Date	Date picker	Date of the treatment
Pesticide	Select	Chosen from the configured pesticide list
Quantity applied	Number	In the unit defined for that pesticide
Target pest / parasite	Text	Free-text description of the pest being treated
Remarks	Text (optional)	Any observations or notes about the treatment

1.4 Harvesting

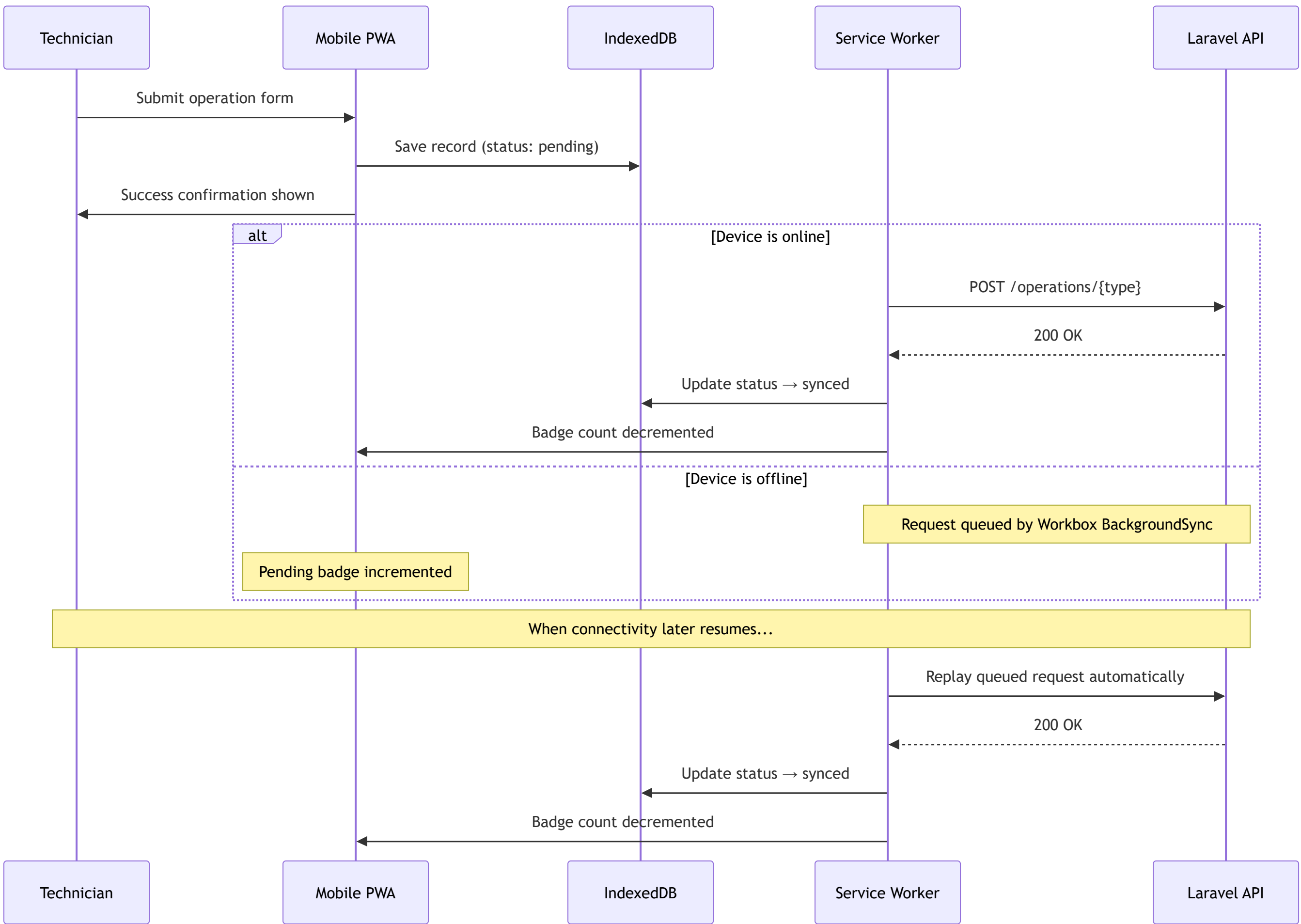
Records a harvest event, capturing labour and yield.

Field	Type	Notes
Plot	Select	Chosen from the configured plot list
Date	Date picker	Date of the harvest
Number of workers	Number	Headcount of workers on that day
Days worked	Number	Duration of the harvest operation (used for cost calculation)
Quantity harvested	Number	Yield in the appropriate unit

2. Offline Behaviour (Mobile PWA)

The mobile app is designed for field use where connectivity may be unreliable.

- **Offline write:** When the technician submits a form without internet, the record is saved locally on the device (IndexedDB) and marked as pending. The technician sees a success confirmation immediately — there is no blocked waiting state.
- **Background sync:** When connectivity is restored, pending records are automatically sent to the server in the background without any action required from the technician.
- **Offline read:** Previously fetched data (plot list, fertilizer list, pesticide list, recent entries) remains accessible from the local cache when offline.
- **Sync status:** A badge in the app header shows the count of unsynced records. A dedicated sync screen lists all pending and errored items, with a manual "Sync now" option and per-item retry.



3. Reporting & Analytics (Manager & Administrator — Web Dashboard)

The manager accesses five report pages, each covering a specific operation type. The administrator inherits full access to all reports (as well as all data entry capabilities from the Technician role). All reports are filterable by plot and date range.

3.1 Irrigation Report

Provides a view of water usage per plot over time.

- **Bar chart:** Water quantity per hectare, grouped by month
- **Bar chart:** Cumulative water applied since the plot's season start date
- **Table:** Total water quantity per plot up to the current date

Filters: plot selector, date range

3.2 Fertilization Report

Provides a breakdown of nutrients applied per plot, auto-calculated from the fertilizer compositions recorded at the time of entry.

- **Table (monthly):** Nitrogen (N), Phosphorus (P), and Potassium (K) units per hectare per month — one column per nutrient, one row per month
- **Table (cumulative):** Total NPK units per hectare since the plot's season start date

NPK units are derived automatically: $\text{applied quantity} \times \text{fertilizer N/P/K\%} \div \text{plot surface area (ha)}$. The manager sees nutrient load directly, without needing to look up fertilizer compositions manually.

Filters: plot selector, date range

3.3 Phytosanitary Report

A detailed treatment log per plot, designed for traceability and compliance.

- **Table per plot:** Treatment date, pesticide name, chemical composition, target pest, remarks
- **Per-column keyword filter:** Each column header has a search input that filters the table rows in real time

Filters: plot selector, per-column keyword search

3.4 Harvesting Report

A yield log per plot.

- **Table per plot:** Date, number of workers, days worked, quantity harvested

Filters: plot selector, date range

3.5 Production Cost Report

A financial summary per plot that aggregates all operation costs for a given period.

Cost line	Calculation
Irrigation cost	$\Sigma (\text{water quantity} \times \text{water unit price})$
Fertilization cost	$\Sigma (\text{quantity applied} \times \text{fertilizer price per unit})$
Phytosanitary cost	$\Sigma (\text{quantity applied} \times \text{pesticide price per unit})$
Harvest cost	$\Sigma (\text{worker count} \times \text{days worked} \times \text{daily labor rate})$
Total production cost	Sum of all above

The report presents one row per plot, with a breakdown of each cost category and the grand total.

Filters: plot selector, date range

4. Configuration & Settings (Administrator — Web Dashboard)

The administrator configures the reference data that all other features depend on. This module must be set up before any data entry can begin. The manager has **read-only** access to all configuration data for reference when interpreting reports.

4.1 Plot Management

Defines the farm's plots. Each plot is the core unit of all operations and reports.

Field	Notes
Plot name	Identifier used throughout the app
Surface area (ha)	Used for per-hectare calculations in reports
Crop type	The category of crop grown (e.g. cereal, fruit tree, vegetable)
Variety	The specific variety of the crop
Season start date	The date from which cumulative totals are calculated for this plot

4.2 Fertilizers

Defines the fertilizer products available for selection during data entry.

Field	Notes
Name	Product name as it appears in the selection list
Unit	Measurement unit (e.g. kg, L)
Nitrogen (N%)	Percentage of nitrogen in the product
Phosphorus (P%)	Percentage of phosphorus in the product
Potassium (K%)	Percentage of potassium in the product
Price per unit	Used in the production cost report

4.3 Pesticides

Defines the pesticide products available for selection during data entry.

Field	Notes
Name	Product name
Unit	Measurement unit
Chemical composition	Active ingredient(s) and formulation description
Price per unit	Used in the production cost report

4.4 Irrigation Water Configuration

Sets the global water measurement and pricing parameters.

Field	Notes
Measurement unit	e.g. m³, litres, mm
Price per unit	Used in the production cost report

4.5 Labor Rate Configuration

Sets the daily labor rate used to calculate harvesting costs.

Field	Notes
Daily labor rate	Cost per worker per day

5. User Management (Administrator — Web Dashboard)

The administrator creates and manages all user accounts. There is no self-registration.

Capability	Notes
Create user	Set name, email, password, and role
Edit user	Update name, role, or reset password
Deactivate user	Suspend access without deleting the account
Role assignment	Roles: technician , manager , admin

6. Internationalisation

Both the mobile app and the admin dashboard support **French (FR)** and **English (EN)**.

- Users can switch languages at any time from the settings (mobile) or the header menu (admin dashboard)
- Language preference is persisted between sessions
- All labels, form fields, error messages, chart axes, and table headers are translated
- Date and number formats adapt to the selected locale (e.g. dd/mm/yyyy and decimal commas for French)